NETFLIX DATA ANALYSIS USING PYTHON

INTERNSHIP PROJECT REPORT

Internship Organization: Auspify Technologies

Internship Program: Data Analysis Using Python

Project Title: Netflix Data Analysis Using Python

Internship Duration: 4 Weeks

Internship Mode: Remote

Submitted By: Shaik Subhani

ABSTRACT

This project focuses on analysing the Netflix Movies and TV Shows dataset using Python. The main objective is to understand the distribution of Netflix content, identify missing values, analyse content categories, explore country-wise representation, examine release-year trends, and derive meaningful business insights.

The project uses Python libraries such as Pandas and Matplotlib for data cleaning, data processing, statistical analysis, and visualization.

The dataset contains 8,807 records and 12 columns. Six analytical tasks were performed, including data cleaning, content type analysis, country-wise analysis, release-year trend analysis, rating and genre analysis, and business insights generation.

The results provide an overview of Netflix's content catalogue and demonstrate how data analytics can transform raw data into meaningful information

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1. INTRODUCTION

Data analysis plays an important role in understanding large datasets and identifying useful patterns that support decision-making.

Netflix is a global streaming platform that provides movies, television shows, documentaries, and other entertainment content.

This project analyses the Netflix Movies and TV Shows dataset to understand its content distribution, country-wise representation, genre categories, release-year patterns, and audience ratings.

The project was completed as part of the Data Analysis Using Python internship program offered by Auspify Technologies.

Python was used to clean, process, analyse, and visualize the dataset. The findings were also used to develop business-oriented recommendations.

2. PROJECT OBJECTIVES

The main objectives of this project are:

- Understand the structure of the Netflix dataset.
- Identify and handle missing values.
- Check and remove duplicate records where applicable.
- Analyse the distribution of Movies and TV Shows.
- Identify countries with the highest content representation.
- Examine Netflix content release-year trends.
- Analyse content ratings and genre distribution.
- Create meaningful charts and visualizations.
- Generate business insights and recommendations.

3. DATASET OVERVIEW

The Netflix Movies and TV Shows dataset was used for this project.

Dataset Name: netflix_titles.csv

Dataset Size: 8,807 rows and 12 columns.

The dataset contains information about Netflix titles, including their names, content types, directors, cast members, countries, release years, ratings, durations, genres, and descriptions.

Dataset Columns

Column	Description
show_id	Unique identifier for each title
type	Movie or TV Show
title	Name of the content
director	Director of the title
cast	Actors and performers
country	Country associated with the title
date_added	Date added to Netflix
release_year	Original release year
rating	Content age rating
duration	Movie duration or TV Show seasons
listed_in	Genre or content category
description	Short content description

4. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used to complete the project:

Programming Language

- Python 3.13.5

Development Environment

- Anaconda
- Spyder IDE
- IPython

Python Libraries

- Pandas – Data loading, cleaning, manipulation, and analysis.
- Matplotlib – Data visualization and chart creation.

- OS – File path management and folder creation.

Data Source

- Netflix Movies and TV Shows CSV dataset.

5. TASK 1: NETFLIX DATA CLEANING AND PREPARATION

5.1 Objective

The objective of this task is to clean and prepare the Netflix dataset by identifying missing values, checking duplicate records, handling inconsistent formatting, and exporting a cleaned dataset for further analysis.

5.2 Methodology

The following steps were performed:

1. Imported the Netflix dataset using Pandas.
2. Examined the dataset structure and column information.
3. Identified missing values in each column.
4. Handled missing values in director, cast, country, rating, and duration columns by replacing them with "Not Available".
5. Converted the date_added column into datetime format.
6. Checked for duplicate records.
7. Standardized text formatting by removing unnecessary whitespace.
8. Exported the cleaned dataset into a new CSV file.

5.3 Missing Value Analysis

The original dataset contained the following missing values:

Column	Missing Values
director	2,634
cast	825
country	831
date_added	10
rating	4
duration	3

Total	**4,307**

5.4 Data Cleaning Results

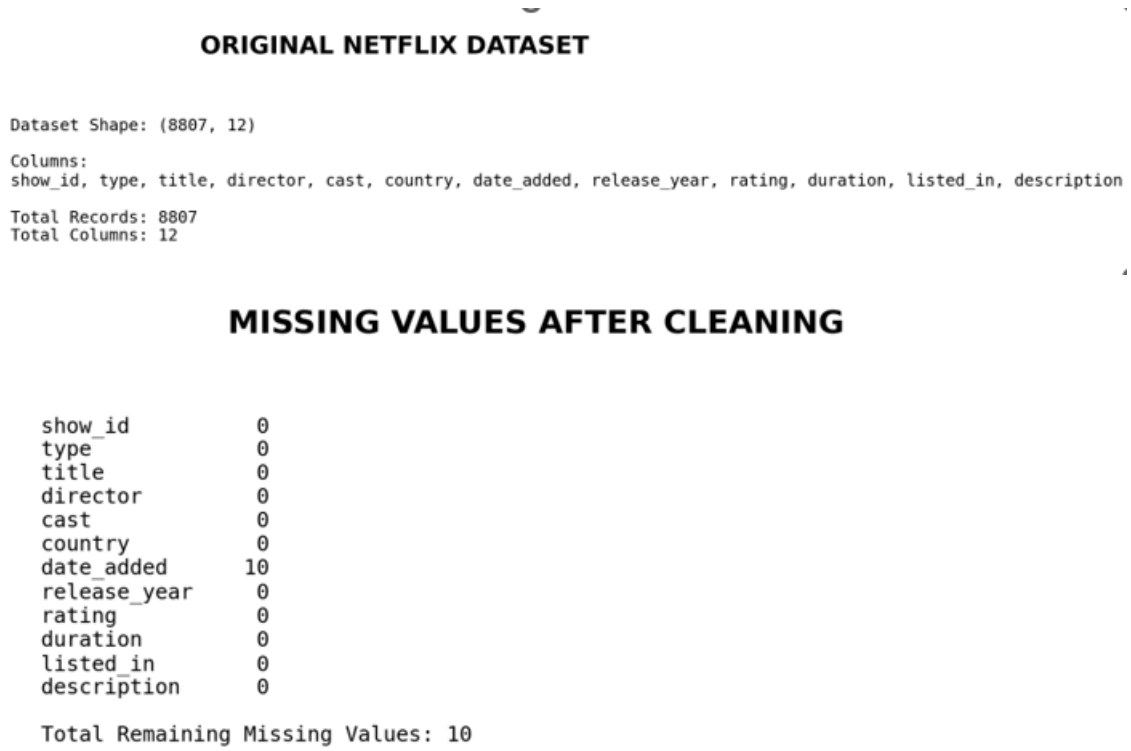
- Original dataset: 8,807 rows and 12 columns.
- Duplicate records identified: 0.
- Duplicate show IDs identified: 0.
- Missing values in director, cast, country, rating, and duration were replaced with "Not Available".
- The date_added column was converted into datetime format.
- Ten unavailable date values remained as missing rather than being assigned invented dates.
- Text formatting inconsistencies were addressed.

The cleaned dataset was exported as:

netflix_cleaned.csv

5.5 Output

The cleaned dataset was successfully saved inside the Task_1_Data_Cleaning folder and used as the input dataset for the remaining analytical tasks.



MISSING VALUES ANALYSIS

```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0

Missing Value Percentage:
show_id          0.00
type             0.00
title            0.00
director        29.91
cast             9.37
country          9.44
date_added       0.11
release_year     0.00
rating           0.05
duration         0.03
listed_in        0.00
description      0.00

Total Missing Values: 4307
```

DUPLICATE CHECK AND TEXT STANDARDIZATION

```
Total Duplicate Rows: 0
Duplicate Show IDs: 0

Text Standardization: Completed
Cleaned Dataset Shape: (8807, 12)
```

CLEANED DATASET VERIFICATION

```
File Saved Successfully!

File Location:
C:\Users\VICTUS\Auspify_Internship\Netflix_Data_Analysis\Task_1_Data_Cleaning\netflix_cleaned.csv

Verified Dataset Shape: (8807, 12)
Remaining Missing Values: 10

Verification Status: SUCCESS
```

5.6 Conclusion

The data cleaning process improved the consistency and usability of the Netflix dataset. The cleaned CSV file provided a reliable foundation for performing further analysis and visualization.

6. TASK 2: CONTENT TYPE ANALYSIS

6.1 Objective

The objective of this task is to analyse the distribution of Movies and TV Shows available in the Netflix dataset and compare their proportions using charts.

6.2 Methodology

The following steps were performed:

1. Loaded the cleaned Netflix dataset.
2. Examined the type column.
3. Counted the number of Movies and TV Shows.
4. Calculated the percentage distribution of each content type.
5. Created bar and pie charts using Matplotlib.

6.3 Content Type Analysis Results

The analysis produced the following results:

Content Type	Number of Titles	Percentage
Movies	6,131	69.62%
TV Shows	2,676	30.38%
Total	**8,807**	**100%**

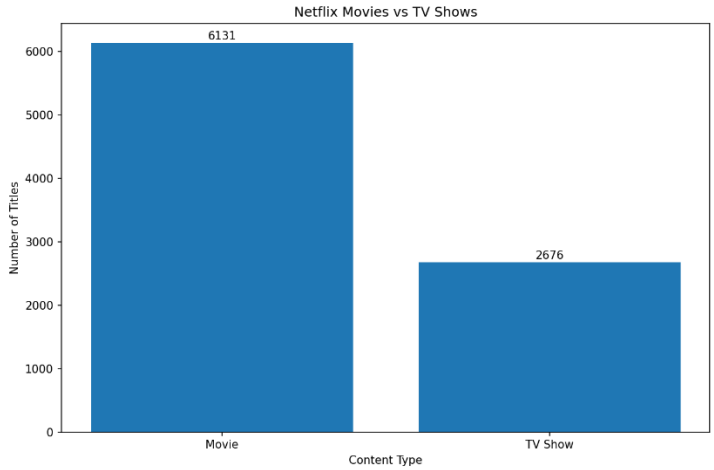
6.4 Key Findings

- Movies represent approximately 69.62% of the total dataset.
- TV Shows represent approximately 30.38%.
- Movies account for the larger share of Netflix titles in the analysed dataset.
- The content catalogue contains both movies and television shows.

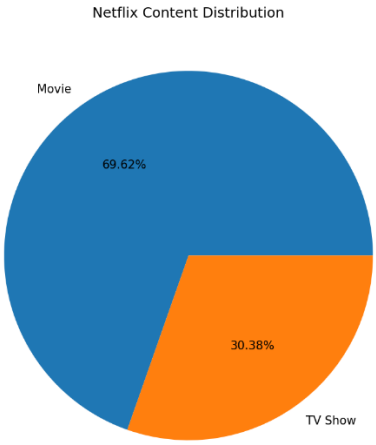
6.5 Data Visualization

The following visualizations were generated:

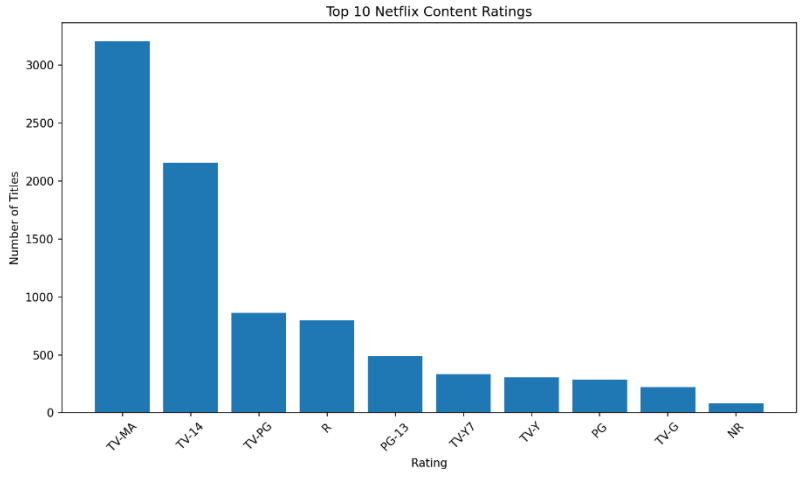
Movie and TV Show Distribution – Bar Chart



Distribution – Pie Chart



Top 10 Netflix Content Ratings



6.6 Conclusion

The content type analysis shows that Movies form the majority of the Netflix catalogue in the dataset. Bar and pie charts helped present the distribution clearly and made it easier to compare Movies and TV Shows.

7. TASK 3: COUNTRY-WISE NETFLIX CONTENT ANALYSIS

7.1 Objective

The objective of this task is to identify the countries with the highest representation in the Netflix dataset and understand the geographical distribution of content.

7.2 Methodology

The following steps were performed:

1. Loaded the cleaned Netflix dataset.
2. Examined the country column.
3. Split multiple country names into individual entries.
4. Removed unnecessary whitespace.
5. Excluded unavailable country information.
6. Counted the individual country credits.
7. Identified the top 10 countries.
8. Created bar and pie charts.

7.3 Country-Wise Analysis Results

The top 10 countries based on individual country credits are:

Rank	Country	Number of Country Credits
1	United States	3,690
2	India	1,046
3	United Kingdom	806
4	Canada	445
5	France	393
6	Japan	318
7	Spain	232
8	South Korea	231
9	Germany	226

10	Mexico	169

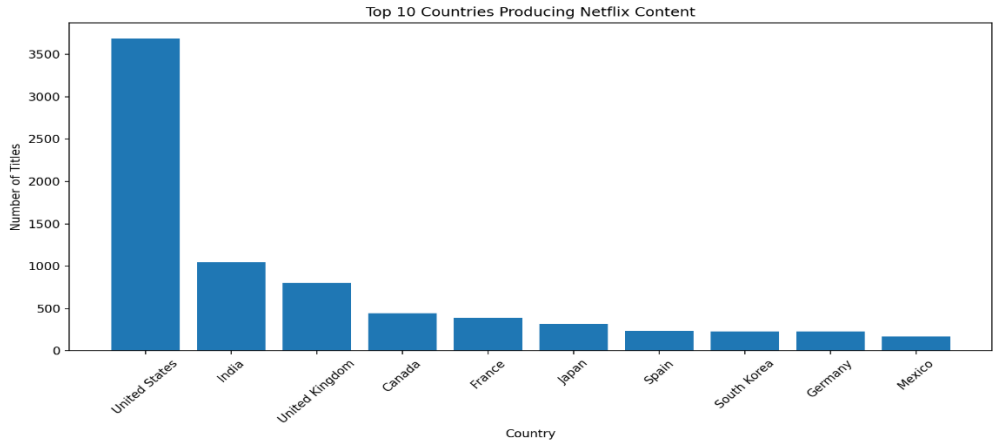
7.4 Key Findings

- The United States has the highest content representation.
- India ranks second in the country-wise analysis.
- The United Kingdom ranks third.
- Canada and France are also among the leading countries.
- The dataset represents content from multiple countries.

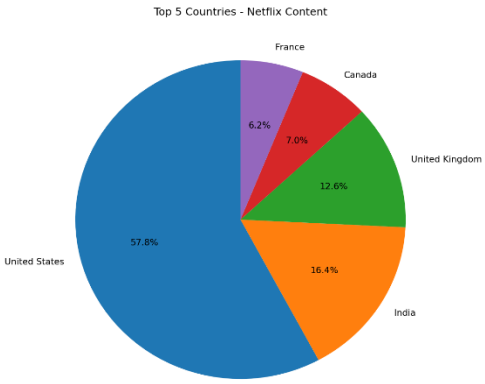
Note: These figures represent country credits, not unique titles. A title associated with multiple countries contributes to each country's count.

7.5 Data Visualization

Top 10 Countries by Netflix Content Credits



Top 5 Countries – Pie Chart



7.6 Conclusion

The country-wise analysis identified the United States, India, and the United Kingdom as the leading countries by content credits. The analysis demonstrates the geographical diversity represented in the Netflix catalogue.

8. TASK 4: TREND ANALYSIS BY RELEASE YEAR

8.1 Objective

The objective of this task is to examine the distribution of Netflix titles across different release years and identify years with higher numbers of released titles.

8.2 Methodology

The following steps were performed:

1. Loaded the cleaned Netflix dataset.
2. Examined the release_year column.
3. Counted the number of titles for each release year.
4. Sorted the yearly counts.
5. Identified the top 10 release years.
6. Compared release-year trends for Movies and TV Shows.
7. Created trend visualizations using Matplotlib.

8.3 Release-Year Analysis Results

The top 10 release years are:

Rank	Release Year	Number of Titles
1	2018	1,147
2	2017	1,032
3	2019	1,030
4	2020	953
5	2016	902
6	2021	592
7	2015	560
8	2014	352
9	2013	288

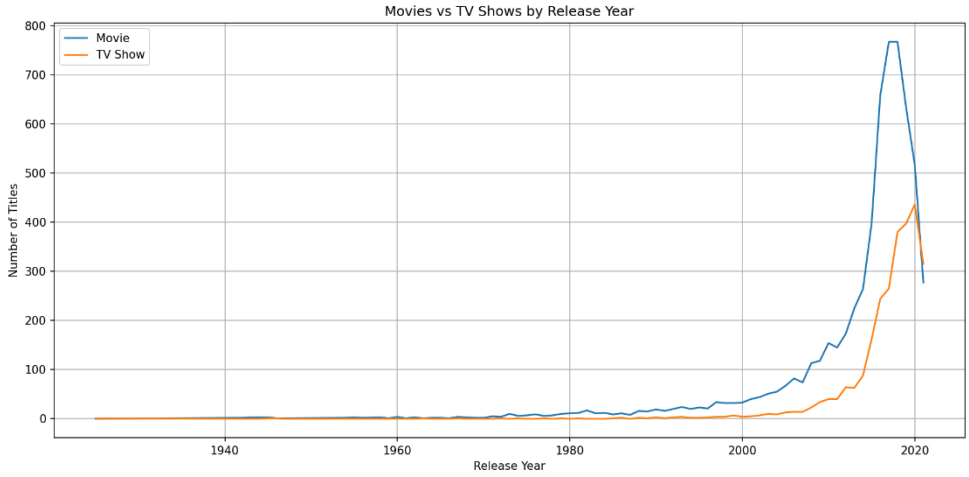
10	2012	237

8.4 Key Findings

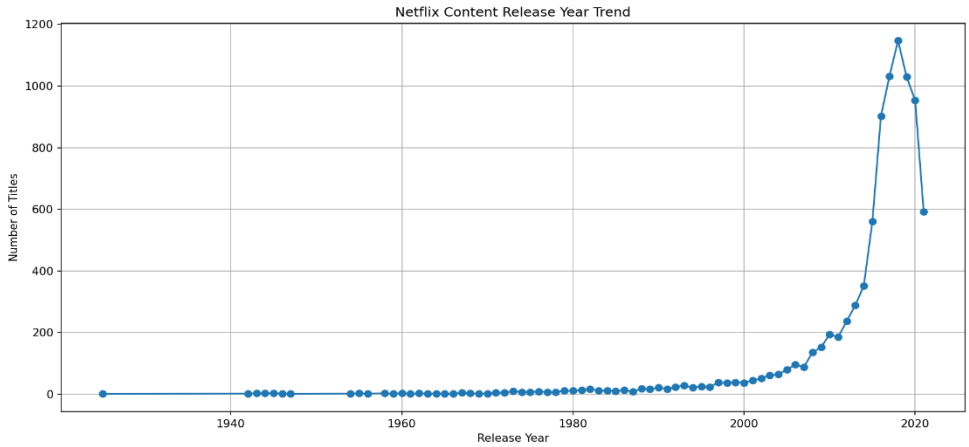
- The year 2018 recorded the highest number of titles, with 1,147.
- The years 2017 and 2019 followed closely.
- The years 2016 to 2020 contributed substantially to the top release-year counts.
- The number of titles in the top release-year rankings declined after 2018.

These figures describe the release years represented in the dataset, not the number of titles added to Netflix during each year.

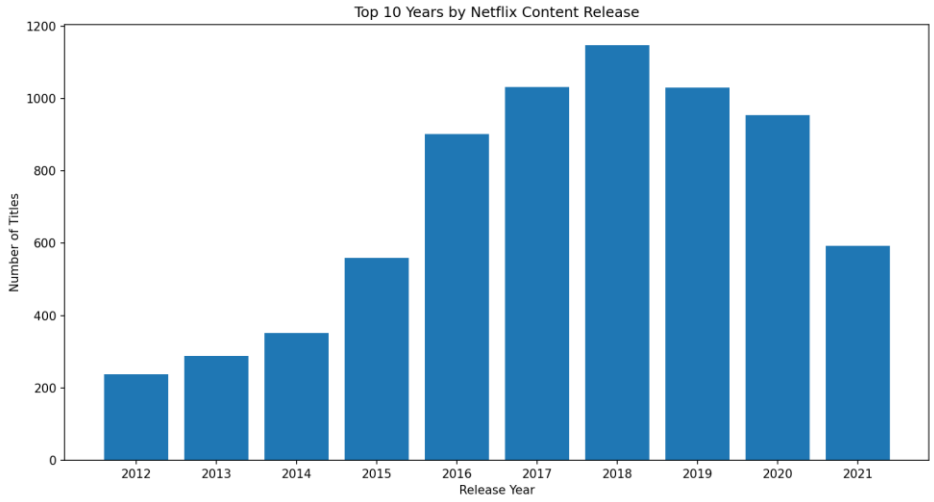
8.5 Data Visualization



Movies vs TV Shows Release-Year Trend



Netflix Titles by Release Year



Top 10 Release Years

8.6 Conclusion

The release-year analysis identified 2018 as the year with the highest number of titles in the dataset. The visualizations helped illustrate the distribution of content across different release years and compare Movies and TV Shows.

9. TASK 5: ADVANCED RATING AND GENRE ANALYSIS

9.1 Objective

The objective of this task is to analyse Netflix content ratings and genre distribution to understand the types of content available and identify the most frequently listed ratings and genres.

9.2 Methodology

The following steps were performed:

1. Loaded the cleaned Netflix dataset.
2. Examined the content rating column.
3. Counted the frequency of each rating.
4. Extracted individual genres from the listed_in column.
5. Split multiple genre values into individual categories.
6. Counted the frequency of each genre.
7. Identified the top 10 ratings and genres.
8. Created bar charts using Matplotlib.

9.3 Rating Analysis

The top 10 content ratings identified were:

Rating	Number of Titles
TV-MA	3,207
TV-14	2,160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80

Key Finding: TV-MA is the most frequently represented rating, followed by TV-14. This indicates a substantial representation of mature and teenage-rated content.

9.4 Genre Analysis

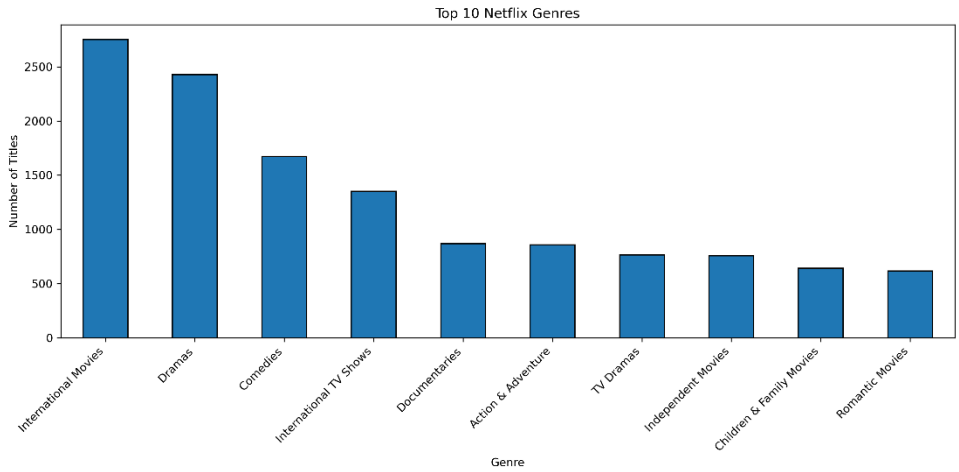
The top 10 genres identified were:

Genre	Number of Genre Credits
International Movies	2,752
Dramas	2,427
Comedies	1,674
International TV Shows	1,351
Documentaries	869
Action & Adventure	859
TV Dramas	763
Independent Movies	756
Children & Family Movies	641
Romantic Movies	616

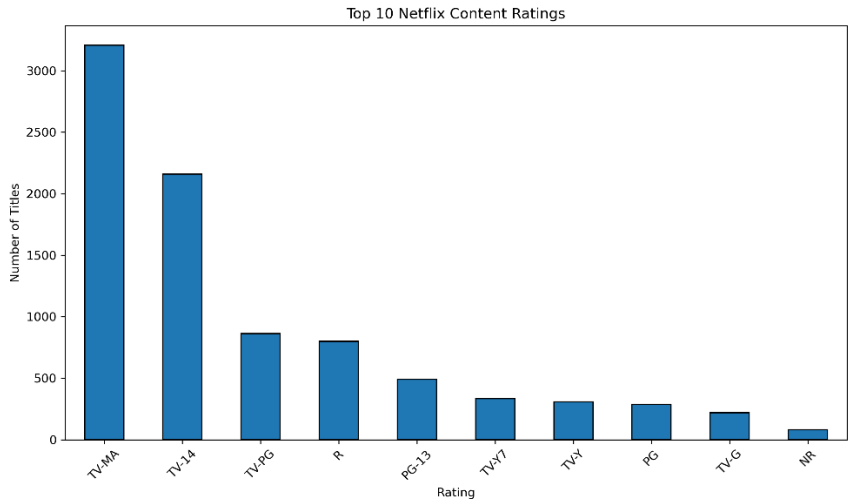
Key Finding: International Movies is the most frequently listed genre, followed by Dramas and Comedies.

9.5 Data Visualization

Top 10 Netflix Genres



Top 10 Netflix Content Ratings



9.6 Conclusion

The rating and genre analysis provided useful insights into Netflix's content distribution. TV-MA was identified as the most common rating, while International Movies was the most frequently listed genre.

The analysis demonstrates how Python can be used to explore content categories and identify patterns in a large dataset.

10. TASK 6: BUSINESS INSIGHTS REPORT

10.1 Objective

The objective of this task is to derive meaningful business insights from the Netflix dataset by combining the findings from content type, country, genre, release-year, and rating analyses.

10.2 Dataset Overview

A total of 8,807 Netflix titles were analysed.

The analysis focused on:

- Movie and TV Show distribution.
- Country-wise content representation.
- Genre distribution.
- Release-year patterns.
- Content rating distribution.

10.3 Content Type Business Insights

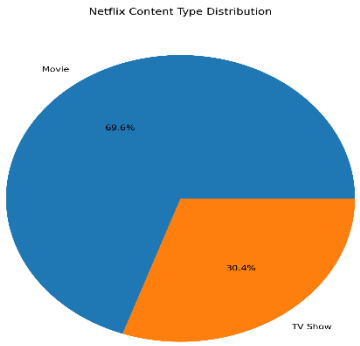
Content Type	Number of Titles	Percentage
Movies	6,131	69.62%
TV Shows	2,676	30.38%

Business Insight:

Movies represent approximately 69.62% of the analysed dataset. This indicates that movies form the larger portion of the available catalogue.

Netflix could evaluate its content mix and consider audience preferences when planning future content development.

Netflix Content Type Distribution



10.4 Country-Wise Business Insights

Country	Number of Country Credits
United States	3,690
India	1,046
United Kingdom	806
Canada	445
France	393

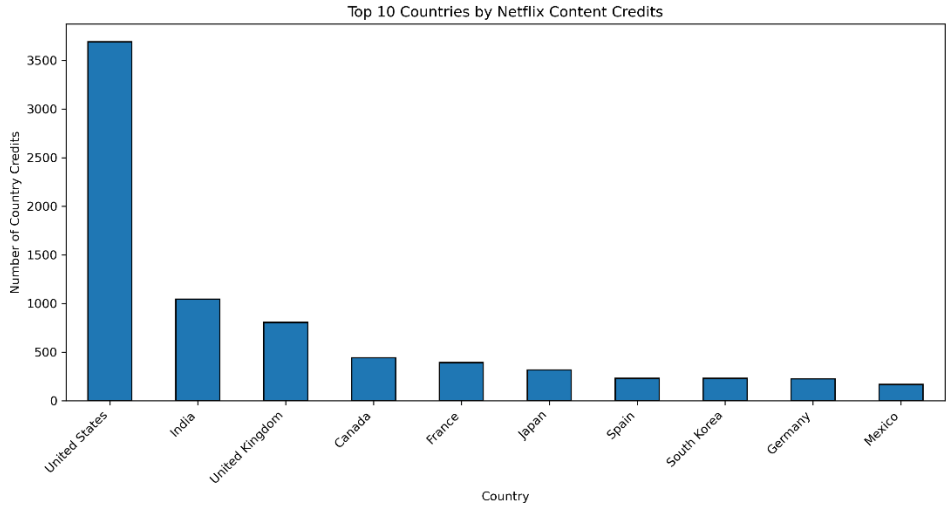
Business Insight:

The United States has the highest content representation, followed by India and the United Kingdom.

This highlights the international distribution of Netflix content and the importance of regional content representation.

Netflix could explore opportunities to strengthen its regional and international content catalogue.

Top 10 Countries by Netflix Content Credits



10.5 Genre Business Insights

Genre	Number of Genre Credits

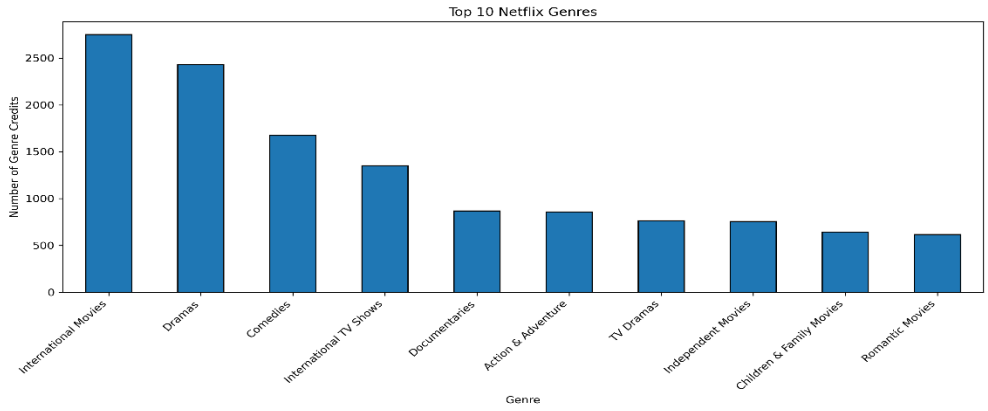
International Movies	2,752
Dramas	2,427
Comedies	1,674
International TV Shows	1,351
Documentaries	869

Business Insight:

International Movies, Dramas, and Comedies are the most frequently represented categories.

These categories can be considered for further content portfolio analysis and strategic planning.

Top 10 Netflix Genres



10.6 Release-Year Business Insights

Release Year	Number of Titles
2018	1,147
2017	1,032
2019	1,030
2020	953
2016	902

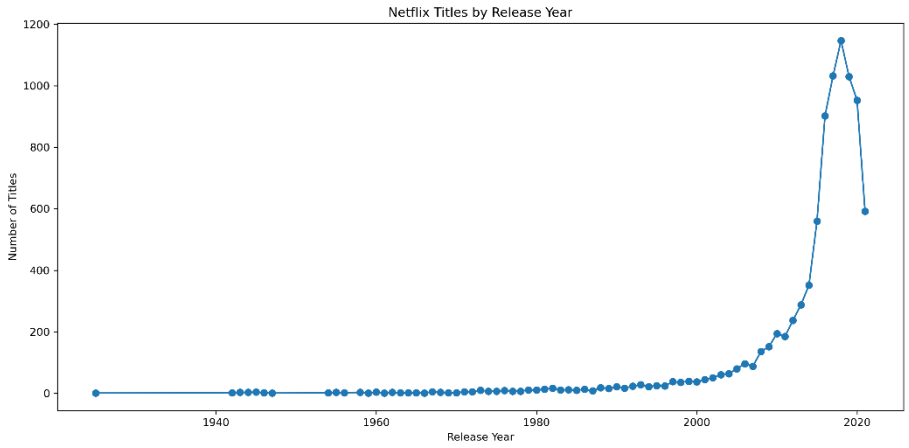
Business Insight:

The year 2018 recorded the highest number of titles in the dataset.

A concentration of titles from 2016 to 2020 is visible among the leading release years.

Monitoring catalogue trends and newer content data can help support future content planning.

Netflix Release-Year Trend



10.7 Content Rating Business Insights

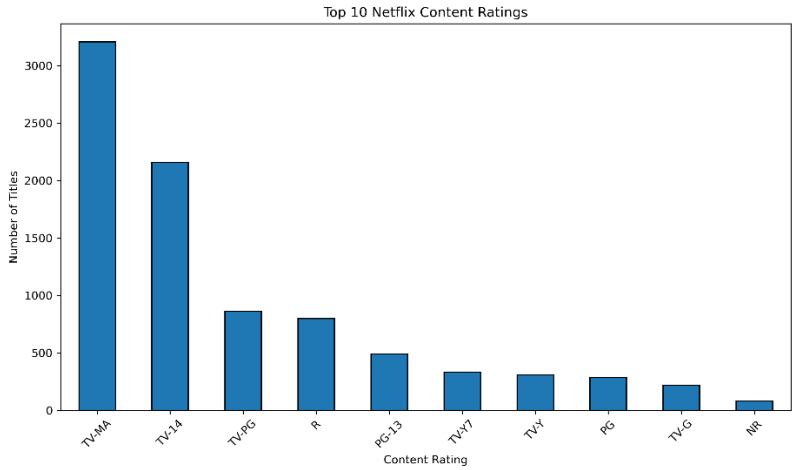
Rating	Number of Titles
TV-MA	3,207
TV-14	2,160
TV-PG	863
R	799
PG-13	490

Business Insight:

TV-MA and TV-14 are the most frequently represented ratings.

Netflix should consider maintaining content variety across different age ratings to serve different audience groups.

Top 10 Netflix Content Ratings



10.8 Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Content Portfolio Management

Review the balance between Movies and TV Shows when planning catalogue development.

2. Regional Content Opportunities

Explore opportunities to strengthen regional and international content representation.

3. Genre-Based Planning

Evaluate frequently represented genres such as International Movies, Dramas, and Comedies.

4. Audience Diversity

Maintain content variety across different age ratings to serve different audience groups.

5. Trend Monitoring

Regularly analyse release-year patterns and updated catalogue data to support future content planning.

10.9 Conclusion

The business insights analysis provided an overview of Netflix's content distribution, country representation, genre categories, release-year patterns, and content ratings.

The findings demonstrate how data analysis can help identify catalogue patterns and support business-oriented recommendations.

11. OVERALL PROJECT FINDINGS

The six tasks completed during this project provided the following major findings:

1. The original Netflix dataset contained 8,807 records and 12 columns.
2. A total of 4,307 missing cells were identified during initial data inspection.
3. No complete duplicate records or duplicate show IDs were found.
4. Movies represent 69.62% of the dataset, while TV Shows represent 30.38%.
5. The United States has the highest country representation, followed by India and the United Kingdom.
6. International Movies is the most frequently listed genre.
7. The year 2018 recorded the highest number of titles by release year.
8. TV-MA is the most frequently represented content rating.
9. Data visualizations helped communicate the findings clearly.

10. Business recommendations were developed based on the analytical results.

12. CHALLENGES FACED

During the project, several practical challenges were addressed:

- Identifying and handling missing values in different columns.
- Processing multiple country names stored in a single column.
- Extracting individual genres from comma-separated values.
- Managing file paths and organizing project folders.
- Creating clear and readable visualizations.
- Interpreting analytical results and converting them into business insights.

These challenges helped improve practical knowledge of Python programming, data manipulation, and analytical thinking.

13. CONCLUSION

The Netflix Data Analysis Using Python project provided practical experience in data cleaning, data processing, exploratory data analysis, visualization, and business insights generation.

All six tasks were completed using Python and its data analysis libraries.

The project began with cleaning the original Netflix dataset and continued with analysing content types, country-wise distribution, release-year trends, ratings, and genres.

The final business insights report demonstrated how analytical findings can be translated into useful recommendations.

This internship project strengthened my understanding of Python, Pandas, Matplotlib, data analysis techniques, and the importance of data-driven decision-making.

Overall, the project provided valuable hands-on experience in working with real-world datasets and presenting analytical results in a structured and professional manner.

14. PROJECT DELIVERABLES

The following deliverables were prepared as part of this internship project:

Deliverable	**Description**
Task 1	Data cleaning and preparation Python script
Task 2	Content type analysis and charts
Task 3	Country-wise analysis and charts

Task 4	Release-year trend analysis and charts
Task 5	Rating and genre analysis with charts
Task 6	Business insights report and charts
Cleaned Dataset	netflix_cleaned.csv
Screenshots	Generated analytical charts
Final Report	Word and PDF internship reports

Project Organization: Each task has a separate folder containing its Python scripts and generated outputs. The Screenshots folder contains the charts created for the report.

END OF REPORT

Submitted to: Auspify Technologies

Submitted by: Shaik Subhani